

Kunal Sharma

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PROFESSIONAL SUMMARY

Senior Data Engineer with 6+ years of experience building **cloud-native Big Data platforms**, data pipelines, and analytics solutions at enterprise scale across **AWS, GCP, and Azure**. Deep expertise in **Apache Spark, Airflow, Snowflake**, and modern cloud data stack tools, with a strong track record of migrating legacy systems to modern architectures. Experienced collaborating across global, cross-functional teams to deliver commercial impact through real-time analytics and scalable data infrastructure. **AWS Certified Solutions Architect**.

WORK EXPERIENCE

S&P Global Jul. 2023 – Present

Senior Software Engineer II – Data Engineering Dallas, TX

- Led the design and delivery of **cloud-native Big Data pipelines** across **AWS, GCP, and Azure**, processing terabytes of financial and ownership data to support Sales, Marketing, and business intelligence objectives.
- Architected and optimized **Apache Spark** workloads (**DataFrames, Spark SQL, Structured Streaming**) for high-throughput batch and streaming data processing, reducing pipeline runtimes by **60%** and enabling near-real-time data access.
- Built end-to-end **Snowflake** data warehousing solutions including schema design, virtual warehouses, Tasks, Streams, and **RBAC**. Implemented **micro-partitioning** and query optimization to reduce costs and improve P99 query latency.
- Migrated legacy ETL pipelines to **DBT** and **Apache Spark**, transitioning from traditional data warehouses and **AWS DMS** to a modern **DataLake** architecture orchestrated via **Apache Airflow (MWAA)**, accelerating data refresh cycles and improving pipeline reliability.
- Integrated **AWS MWAA** with **AWS Bedrock** to build intelligent **Airflow** DAGs that surface pipeline anomalies and AI-generated resolutions to engineering teams, reducing incident response time by **40%**.
- Developed **REST APIs** and data integrations to expose analytical datasets to Sales and Marketing tools, reducing ad hoc requests by **50%**. Deployed **GitHub Copilot** agents for automated code validation and vulnerability detection, reinforcing secure coding standards across the team.
- Managed multiple data warehouses (**Snowflake, SQL Server, PostgreSQL**) and oversaw data refresh strategies, ensuring consistent and timely updates from heterogeneous source systems. Administered an **Alteryx Server** on **AWS ECS** with a containerized **MongoDB** store for user metadata and an **RDS** backend for workflow persistence, enabling self-serve analytics for business stakeholders.
- **Stack:** Python, PySpark, Apache Spark, Snowflake, AWS (S3, EC2, Lambda, Glue, EMR, Step Functions, MWAA, DynamoDB, Athena, Redshift, SageMaker, Bedrock), GCP (BigQuery, Dataproc, Dataflow), Azure DevOps, Terraform, DBT, Airflow, Docker, GitHub Actions, REST APIs

Cognizant Aug. 2022 – Jul. 2023

Senior Associate – Data Science Engineer Dallas, TX

- Designed cloud-based data warehouse architectures leveraging **AWS (S3, DynamoDB, EC2, Glue)** and **Snowflake** to drive BI initiatives for financial clients, reducing report generation time by **35%**.
- Built **Apache Spark** applications in **Python** for large-scale data transformation, cleaning, and validation. Developed batch and streaming ingestion pipelines using **Hadoop, Hive, Sqoop, and MapReduce**.
- Implemented real-time streaming applications with **Apache Kafka** and **Apache Spark** to enable low-latency analytics on high-volume event data for client-facing reporting.
- Automated infrastructure provisioning with **Jenkins CI/CD** and **GitHub** version control. Migrated data from **Redshift** to **SQL Server** and coordinated **Tableau** dashboard delivery for reporting stakeholders.
- Collaborated with cross-functional global teams using Agile/Scrum, JIRA, and Confluence to deliver data products aligned with evolving business requirements.

Kent State University Jan. 2022 – Aug. 2022

Data Analyst Kent, OH

- Designed and deployed financial dashboards in **Tableau** and **Power BI**, providing real-time ROI visibility across fundraising events. Delivered complex analytical reports to stakeholders for KPI measurement.
- Built ML classification and regression models (**SVM, Random Forest, Decision Tree**) in **Python** to predict event ROI. Standardized data science workflows under **Git** version control.

Wipro Technologies Aug. 2018 – Dec. 2020

Project Data Engineer India

- Developed and maintained **ETL pipelines** using **SQL/PLSQL, Python**, and **Shell Scripting** for a large telecom client, supporting billing, MNP processing, and churn analytics at scale.
- Built a customer **churn prediction model** using **Python** and **Random Forest** with association rules, enabling targeted retention campaigns and measurably reducing customer churn.
- Increased customer satisfaction by **32%** through CRM-driven initiatives. Improved **SLA/OLA compliance by 50%** within two quarters by delivering production-stable billing support applications.

EDUCATION

Kent State University May 2022

Master of Science, Business Analytics

APJ Abdul Kalam Technical University May 2018

Bachelor of Technology, Computer Science and Engineering

CERTIFICATIONS AND SKILLS

- **Certifications:** AWS Certified Solutions Architect – Associate
- **Big Data and Processing:** Apache Spark (DataFrames, Spark SQL, Structured Streaming), Apache Kafka, Hadoop, Hive, Airflow (MWAA), AWS Glue, DBT, Flink (familiar)
- **Cloud and Infrastructure:** AWS (EC2, S3, IAM, VPC, Lambda, ECS/EKS, Glue, Step Functions, CloudWatch, EMR, Bedrock, SageMaker) | GCP (BigQuery, Dataproc, Dataflow) | Azure (DevOps, Data Factory) | Terraform | Docker
- **Databases (NoSQL and Relational):** Snowflake (data warehousing, virtual warehouses, Tasks, Streams, RBAC, query optimization), PostgreSQL, SQL Server, Oracle DB2, MongoDB, DynamoDB, Redshift, Athena, Neo4j — in-depth SQL and columnar DB design
- **Languages and APIs:** Python, SQL, PySpark, Scala (familiar), REST APIs, GraphQL (familiar), Shell Scripting
- **ML and AI:** Scikit-learn, TensorFlow, PyTorch, Keras, NLP, CNN, SVM, Random Forest, LLM, AWS SageMaker, AWS Bedrock, GitHub Copilot
- **DevOps and Tools:** GitLab CI/CD, GitHub Actions, Jenkins, CloudWatch, Grafana, Prometheus, OpenTelemetry, JIRA, Confluence, Liquibase, Tableau, Power BI
- **Methodologies:** Agile/Scrum, SDLC, Microservices, Event-Driven Architecture, Domain-Driven Design